ITUH USED HS-11-2020PM PAP

R13

Code No: 117BX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, October/November - 2020
COMPUTER GRAPHICS
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Define aspect ratio and resolution. How are they useful in Computer Graphics?
b) Illustrate various input and output devices in detail. [7+8]

2. Write down and explain the midpoint circle drawing algorithm. Assuming 10 cm as the radius and origin as the centre of the circle explain the algorithm. [15]

3. Show that a reflection about the line y = -x is equivalent to a reflection relative to y-axis followed by a counter clockwise rotation of 90^0. [15]

4. Explain in detail the Cohen-Sutherland line clipping algorithm with an example. [15]

5. Suppose we have a B spline curve of degree 3with a knot vector as follows
U0-to-u3 -0; u4-0.2; u5-0.4; u6-0.6;u7-0.8; u8-to-u11-1
Insert a new knot t=0.5, find new control points and new knot vector. [15]

6. Perform a perspective Projection of the unit cube onto the x = 0 plane from the center of Projection on x-axis at xc = 10. [15]

7. Explain the depth buffer method of detecting the visible surfaces? [15]

8. Explain the steps involved in the design of animation sequence. [15]

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